Hassan Imani

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PROFESSIONAL EXPERIENCE

Machine Learning Engineer | AML AI Research & Special Projects — Jul 2025 – Present
Scotiabank — *Toronto, ON*

- **Generative Modeling & Anomaly Detection:** Refactored and optimized conditional generative anomaly detection models conditioned on KYC descriptors, isolating peer-group outlier behavior to suppress false positives in high-volume transaction streams.
- **Foundational Representation Learning:** Built self-supervised sequence models on temporal transaction logs to produce dense client embeddings; established evaluation suites measuring semantic clustering against demographic and risk cohorts.
- **ML Platform & Synthetic Benchmarking:** Developed a custom Hyperparameter Optimization (HPO) tool to traverse parameter spaces and designed synthetic transactional benchmarks to unblock delayed-feedback validation loops.
- **Agentic Alert Investigation Pipeline:** Contributed to an agentic narrative generation workflow for AML compliance; ingests multi-entity customer context (KYC profile, account records, transaction velocity, counterparty networks), enforces regulatory typology guardrails, and evaluates red-flag criteria to synthesize standardized case narratives.
- **Automated Evidence Synthesis & Delivery:** Engineered a batch orchestration loop that formats synthesized case narratives, customer behavioral profiles, and transactional evidence into structured workbooks for compliance investigator review and case resolution.

Machine Learning Engineer — Feb 2024 – Jul 2025
Zown — *Toronto, ON*

- **Listing Recommendation Engine:** Architected a multi-stage candidate matching engine pairing deterministic constraint filtering with a Two-Tower neural retrieval network, integrating multi-task MLP heads predicting dwell time and interaction click-throughs.
- **Multi-Party Conversational Agent (Offer Bot):** Contributed to an end-to-end agent in WhatsApp group chats orchestrating transactions among buyers, licensed agents, and AI; parsed listing IDs, triggered automated property valuation fallbacks, and dynamically retrieved disclosure documents from vendor databases.
- **Tool Calling & Document Automation:** Implemented programmatic tool-use workflows to map, extract, and populate standardized legal contracts (OREA Forms 100, 801, 320) via coordinate-based form-field APIs, integrating DocuSign eSignature to generate live execution envelopes in chat.
- **Continuous Training & Validation Guardrails:** Deployed tabular property valuation models with an automated Continuous Training (CT) pipeline and holdout validation gates to prevent production metric degradation.
- **Multimodal Content Understanding:** In-housed listing photo understanding by fine-tuning Vision Transformers (Google ViT) with multi-task classification and regression heads to supply structured visual signals to downstream ranking models.

EDUCATION

York University — 2021 – 2025
PhD in Mechanical Engineering — *Toronto, ON*

- Equivariant Graph Neural Networks (eGNNs), domain adaptation, spatio-temporal tracking, and statistical regression modeling.

Tehran Polytechnic — 2018 – 2021
MSc in Mechanical Engineering — *Tehran, IR*

Sharif University of Technology — 2013 – 2018
BSc in Mechanical Engineering — *Tehran, IR*

TECHNICAL SKILLS

AI Agents & Workflows: Multi-Step Agents, Tool Use & API Integration, Guardrails & Evaluation Suites

ML & Modeling: Anomaly Detection, Two-Tower Recommenders, Representation Learning, GBDTs / Classification, GNNs, ViTs

Frameworks & Platforms: PyTorch, PyTorch Geometric, Scikit-Learn, PySpark, Spark, SQL, Docker, GCP, Git

Languages: Python, SQL, Bash

PUBLICATIONS

Graph Neural Networks for Sustainable Energy (HI Parashkooh, C Jian) — 2025
Predicting Adsorption in Aromatic Molecules — *ChemE 9 (4), 85*

An Innovative Method for Calculating PMV (HI Parashkooh, C Jian) — 2024
from a standard molecular trajectory — *J. Mol. Liq. 395, 123879*

Data Mining Guided Molecular Investigations (HI Parashkooh, C Jian) — 2022
on the Coalescence of Water-in-Oil Droplets — *Energy Fuels 36 (4), 1811*